

Assumptions

The analysis makes the following assumptions:

1. Brent crude oil prices accurately represent global crude oil market conditions.
2. Event dates are treated as the beginning of market reactions, although markets may respond before or after official announcements.
3. Each major geopolitical or economic event is assumed to have the potential to influence oil prices independently.
4. Daily closing prices sufficiently capture market responses.
5. External factors not included in the event dataset (weather, currency fluctuations, speculation, technological changes, etc.) are assumed to have smaller or random effects.

Limitations

Several limitations should be acknowledged.

Correlation vs Causation

The most important limitation is that statistical analysis identifies **temporal associations**, not definitive causal relationships.

A change point occurring shortly after a geopolitical event does **not** prove that the event caused the price change. Other unobserved factors may have influenced the market simultaneously.

Therefore, findings should be interpreted as evidence of **statistical correlation**, supported by economic reasoning, rather than proof of causality.

Other Limitations

- Markets often react before official announcements due to expectations and leaked information.
- Multiple events can occur simultaneously, making it difficult to isolate individual effects.
- Some geopolitical events unfold gradually rather than occurring on a single date.
- Oil prices are also influenced by exchange rates, inflation, speculation, technological advances, and macroeconomic conditions that are not explicitly modeled.
- Bayesian Change Point Analysis detects statistical shifts but does not identify the underlying cause of those shifts.